

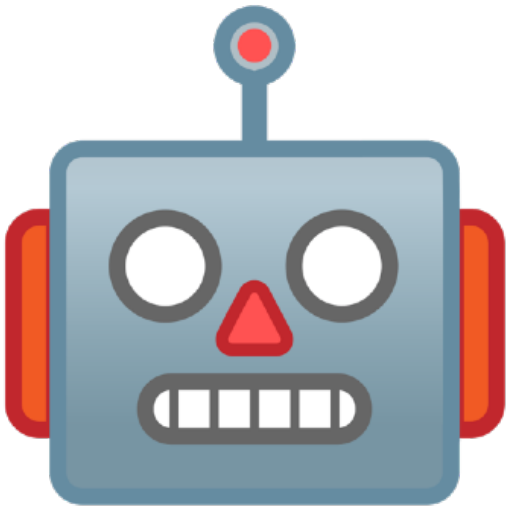
Our goal is to compare these two modalities.

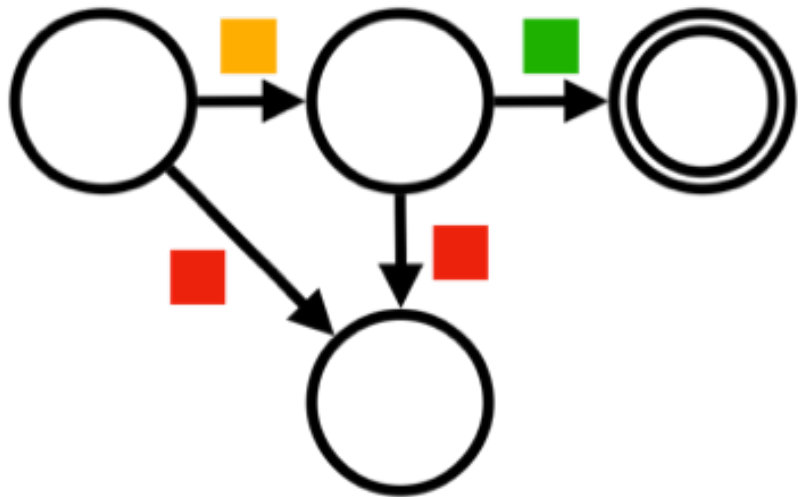
1. Which modality is better for learning-based control?

2. What happens when these modalities are combined?

Challenge: How to generate authentic instruction pairs?

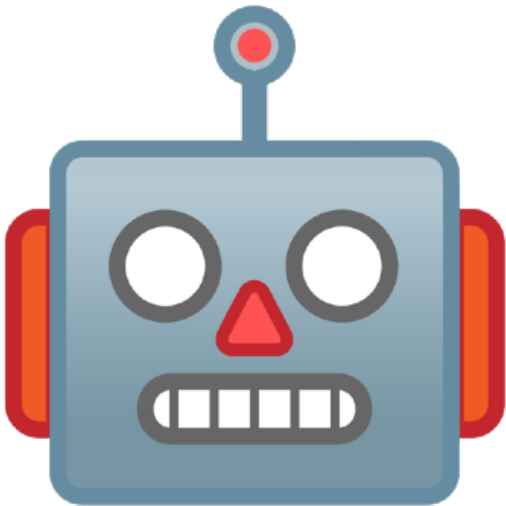








Pick up the laundry
and fold it. Avoid
breaking anything!





RAD *Embeddings*



Qwen3-Embedding







Autonata)

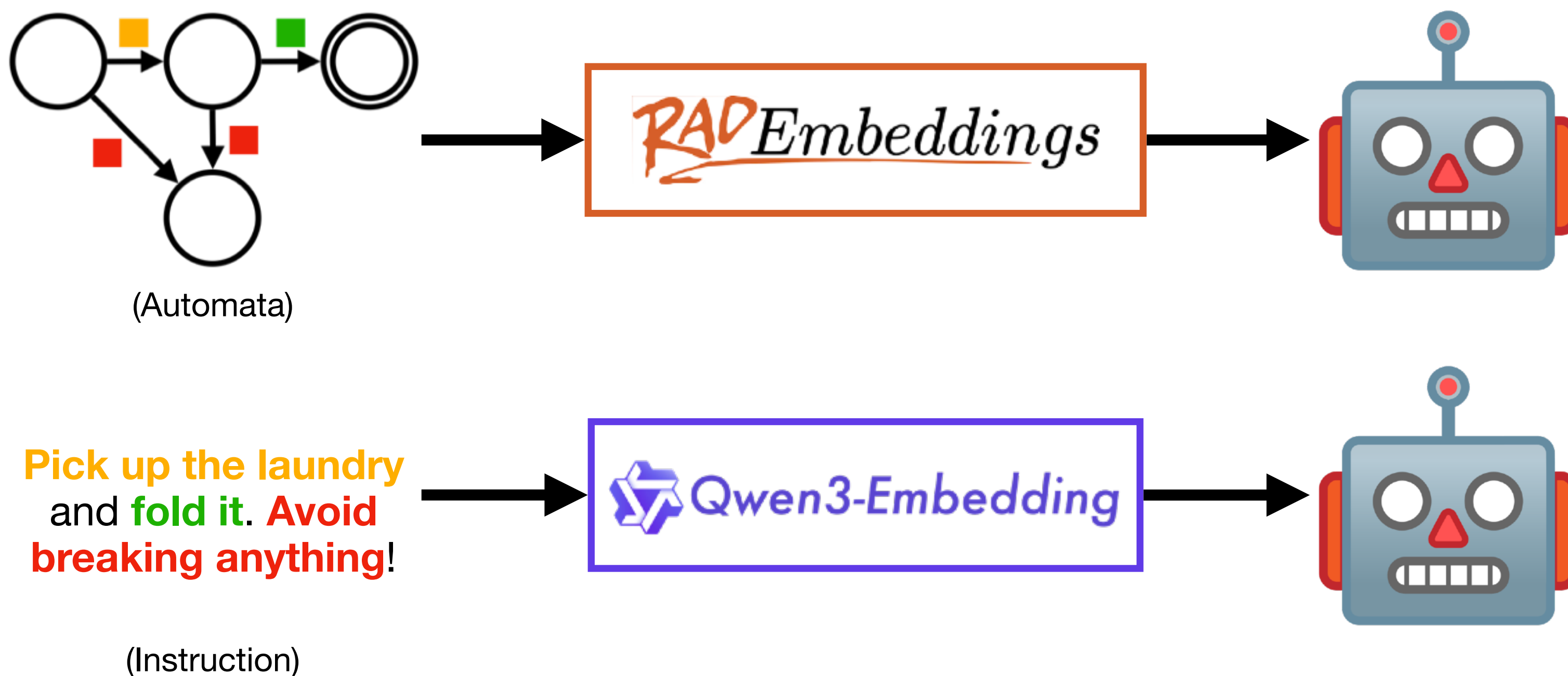
instruction)

Yalcinkaya, Beyazit, et al. "Compositional automata embeddings for goal-conditioned reinforcement learning." Neurips 2024.

Zhang, Yanzhao, et al. "Qwen3 embedding: Advancing text embedding and reranking through foundation models." arXiv 2025.



Our goal is to compare these two modalities.



1. Which modality is better for learning-based control?
 2. What happens when these modalities are combined?
- **Challenge:** How to generate automaton-instruction pairs?

Outline

- ~~Introduction~~
- **Preliminaries**
- Synthesizing Language Instructions from Automata
- Learning Policies with Automata and Instruction Embeddings
- Empirical Study
- Discussion